

KiCAD Workshop Plan

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Goal: Teach students how to use free PCB design software to order custom circuit boards for projects

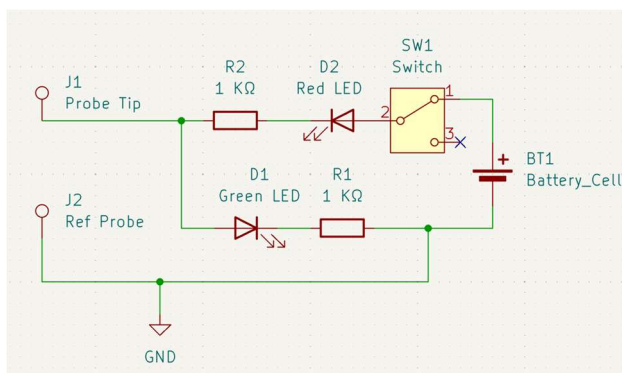
- First in series of workshops on engineering skills including soldering the parts in place and Solidworks modeling and 3D printing a case to hold the PCB

Time: 45 mins

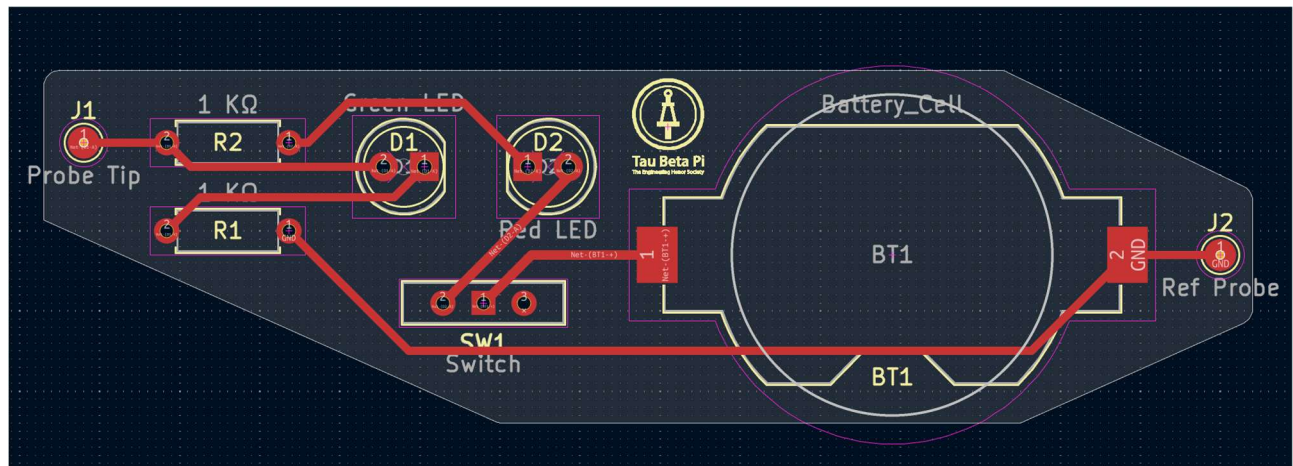
Design Concept

- Pocket-sized continuity tester
 - Diagnostic tool for testing circuits. Touching the probe tip to a circuit node turns on the Green LED for HIGH (3V-5V) or the Red LED for LOW/GND.
 - **Component List:**
 - 1× CR2032 coin cell holder
 - 1× 5mm Green LED (HIGH indicator)
 - 1× 5mm Red LED (LOW indicator)
 - 2× 1K resistors
 - 2× Male header pins (acting as test probes)
 - Easy to construct schematic and visualize

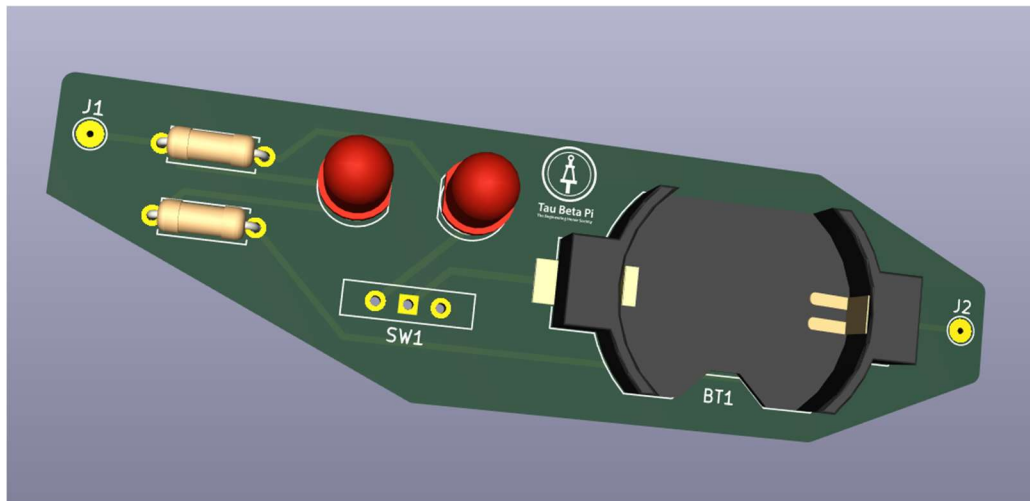
Completed Schematic:



PCB Design:



3D Model:



Reference	Component	Symbol	Footprint
BT1	CR2032 3V Battery Holder	Battery_Cell	BatteryHolder_Keystone_1058_1x2032
SW1	SPDT Slide Switch	SW_SPDT	SW_Slide-03_Wuerth-WS-SLTV_10x2.5x6.4_P2.54mm
D1, D2	Green, Red LED	LED	LED_D5.0mm
R1, R2	1KΩ Resistor	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal

TP1, TP2	Signal Probe Tip, GND Ref Terminal	TestPoint	TestPoint_Pad_D2.0mm
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Preliminary

- Ahead of time, make sure students have KiCAD downloaded to their computer and open the app to make sure it works
- Read **Getting Started in KiCAD** up to and including **First Time Setup**
 - Available on website:
https://docs.kicad.org/10.0/en/getting_started_in_kicad/getting_started_in_kicad.html#introduction-to-kicad

Handout Reference Sheet

Section 1: Introduction (5-10 mins)

Hey everyone, welcome to our first Tau Beta Pi KiCAD workshop. I'm Anthony, and I'll be helping you along as we make a custom continuity tester.

Everyone please open KiCAD or quickly download it from the official website.

1. Share screen as I act so everyone can follow along

I'll start by showing you what a completed PCB looks like.

2. Open Example Board (Adafruit Feather) -> 3D model

If you're not familiar, a Printed Circuit Board, or PCB, is a non-conductive board layered with conductive copper tracks that route electrical signals. It mechanically holds and electrically interconnects electronic components that you connect to it—such as microchips, resistors, and capacitors—replacing bulky wiring systems. All modern electronics are made using PCBs, including medical devices. Think of it as the tool separating real devices from Dr. Lippmann class projects with breadboards and jumper cables.

This project is going to be a fairly simple and practical continuity tester that you can use to examine circuits that you work on in your own projects.

1. Open Schematic with Symbols

We'll start by making a **Schematic** that shows where the electrical connections will take place. These are like the circuit drawings you use in Physics II with resistors and capacitors and stuff, with those being **Symbols** that we're going to use to represent the actual components we're using.

2. Open PCB Editor with Footprints

After that, we design the board physically using the **PCB Editor**. Now, the electrical symbols turn into **Footprints**, which represent the physical space that component takes up on the board.

Section 2: Schematic (15 mins)

1. Create new project and open schematic

Now go under File, New Project, Default, OK, and name it whatever you'd like.

Next, open up the Schematic Editor

2. Place and wire components (from default library) following the template

Now we will start placing Symbols (Press Place or the A key). KiCAD has a default library which has all of the symbols and footprints we are going to use. Look at the table at the bottom of your reference sheet and use the symbol name there to place each one into the schematic.

Draw wires between symbols using the wire tool on the right here or pressing the W key.

3. Place No Connect Flag [Q] over switch pin 3

4. Validate using ERC

To make sure our circuit doesn't have any fatal flaws, press Inspect, then run the Electrical Rules Checker. It will say there's an error on the GND pin, but that's not going to apply to us.

5. Assign footprints to each component (also from default library)

Now the last thing we do in the Schematic Editor is to assign Footprints to each Symbol. That way, they will transfer automatically to the PCB Editor. Double-click each symbol and go to the Footprint row of the table under the Value column. Press the library symbol in that column and then look up the Footprint next to that symbol in the table on your reference sheet. Press OK, and then repeat it for every symbol except GND.

[Pause to take questions and assist students who need help]

Section 3: PCB Editor (15 mins)

Now we press PCB Editor in the top right corner. This is the space we will be using to organize where the components will actually go.

1. Update PCB from schematic

Then press Update PCB from schematic, and when we do that, we're given a bunch of Footprints we can place and move around.

Before you do anything, we need to go to board setup in the top left and under Net Classes, change default track width to 0.5mm.

2. Arrange components and route tracks

Now, we can finally start arranging components and routing tracks between them. Since we made the schematic first, you can see there are already lines telling us which connection points need to be made.

3. Explain PCB layers (Copper trace layers, GND plane, Silkscreen, etc)

On the right, you can see all the layers that make up the board. Most of them we don't need to worry about, but what's important is the Red one (the front copper layer that we're making all our connections on), the Blue one (the back copper layer which we can use if we have tracks that overlap), the Silkscreen layer (where you add text and designs so that they show up on the final board), and the Edge Cuts layer (where you draw the actual shape of the board you want printed).

4. Draw board outline and mounting holes (if needed) in Edge.Cuts

We'll draw that now by clicking on that layer and drawing lines from one point to the next.

5. Footprint Editor -> File -> Import -> Footprint -> TBP Logo Footprint Folder

For an example of how the Silkscreen layer can be used, I'm importing a premade graphic to decorate my board. If you want, you can stay after and I could show you how to add any image or custom design to your PCB.

6. Verify using DRC and 3D viewer

Last, we verify the board by running the Design Rules Checker. Once that is sorted out, we use the 3D Viewer on the top ribbon to see what our PCB is going to look like in real life.

[Pause to take questions and assist students who need help]

Section 4: Next Steps (10 mins)

1. Show how to export Gerber files and send to PCB manufacturer

We can send what we just made to a manufacturer using a Gerber Plot. Go under File -> Fabrication Outputs -> Gerbers. You can then send that file to a manufacturer and order your PCB to be made exactly the way you designed it.

You can also export your design as a STEP file, which you can use in SolidWorks assemblies.

If time allows

1. Use silkscreen layer and/or image converter to add designs and graphics
 - a. Download image
 - b. Open Image Converter
 - c. Import image
 - d. Tweak output size to dimensions appropriate for your board
 - e. Select "Footprint"
 - f. Export to File
 - g. Open Footprint Editor
 - h. Import Footprint
 - i. Save to New Library
2. Do extra GND layer with Filled Zones

3. Highlight physical design choices that support soldering (Thermal reliefs, pads)

Custom PCB Fabrication

- **Design Specs for Board House Order (JLCPCB, PCBWay, or OSH Park):**
 - **Layer Count:** 2-Layer
 - **Material / Thickness:** FR-4, 1.6 mm standard thickness
 - **Copper Weight:** 1 oz / 35um
 - **Surface Finish:** HASL (Lead-Free) or ENIG
 - **Soldermask Color:** Green, Blue, or Black (high-contrast background for silkscreen art)
 - **Silkscreen:** White text/art
- Workshop Hardware & Consumables (Group / Lab Use)
 - **Test Leads:** 1× Alligator-to-Alligator Clip Wire per student (for clipping J2 to circuit ground).
 - **Probe Needle Options:** 1× 1.5" piece of 20 AWG or 22 AWG solid copper wire per student (to solder into J1 as a rigid probe point).
 - **Keyring Attachments:** 25mm split rings (if utilizing a mounting hole).

Procurement Summary for a 20-Student Cohort (With 15% Spares)

- **Resistors (1K Ohm, 1/4W):** 50 pcs (buy a 100-pack cut tape)
- **5mm Green LEDs:** 25 pcs
- **5mm Red LEDs:** 25 pcs
- **SPDT Slide Switches (EG1218):** 25 pcs
- **CR2032 Holders (MPD BS-7):** 25 pcs
- **CR2032 Batteries:** 25 pcs
- **0.1" Male Pin Header Strips:** 2× 40-pin header strips (broken into single pins)
- **Custom PCBs:** Order 25 or 30 boards (minimum batch size is usually 5 or 30 pcs depending on supplier)
- **Alligator Clip Jumpers:** 1 pack of 10–20 test leads